

API - Documentation

Here we offer an API (Application Programming Interface) to assist users in integrating DDMut-PPI into their research pipelines.

In summary, all jobs submitted to our server are labelled with a unique ID which is used to query the status (in queue, processing, processed).

DDMut-PPI allows for six distinct types of submissions:

- [Single Prediction](#)
- [List Prediction](#)
- [Alanine Scanning](#)
- [Saturation Mutagenesis](#)
- [Multiple Mutations: Manual Prediction](#)
- [Multiple Mutations: Systematic Evaluation](#)

Single Prediction -

http://biosig.lab.uq.edu.au/ddmut_ppi/api/single

POST - Job Submission

Arguments

- **pdb_accession** (optional) - 4 character PDB code
- **pdb_file** (optional) - file in [PDB format](#)
- **chain** (required) - Chain identifier
- **mutation**(required) - Point mutation code (aaFrom + residueNumber + aaTo)
- **Reverse** (optional) - Whether to predict reverse mutations (True/False). Default is False
- **email** (optional) - Email for contact when the job is finished

Return

- **job_id** - ID used for uniquely identify each job

Examples

- **curl**
\$ `https://biosig.lab.uq.edu.au/ddmut_ppi/api/single` `curl -X`

```
POST -i -F pdb_file=@/home/ubuntu/1cse.pdb -F
mutation=L45G -F chain=I -F reverse=True
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 38
Date: Tue, 21 Nov 2023 04:36:23 GMT
{
  "job_id": "17005413833647785"
}
```

GET - Retrieve Job Results

Arguments

- **job_id** - ID used for uniquely identify each job. Generated upon submission

Return

For jobs still being processed or waiting on queue, the message below will be returned from querying this endpoint:

≡ DDMut-PPI

For jobs successfully processed by DDMut-PPI, the following data will be returned:

[Data](#) [Help](#) [API](#) [Privacy](#) [Contact](#) [Acknowledgements](#)

- **prediction** - Related Resources

Examples

- **curl**

```
$ curl -X
https://biosig.lab.uq.edu.au/ddmut_ppi/api/single -X
GET -F job_id=17005413833647785
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 33
Date: Wed, 10 Jun 2022 05:10:59 GMT
{
  "job_id": "17005413833647785",
  "status": "DONE",
  "prediction": -3.323,
  "prediction_reverse": 2.905,
  "chain": "I",
```

```
"position": 45,  
"wild-type": "LEU",  
"mutant": "GLY",  
"distance_to_interface": 2.72,  
"results_page":  
"https://biosig.lab.uq.edu.au/ddmut_ppi/results_prediction/17005413833647785",  
}
```

List Prediction -

http://biosig.lab.uq.edu.au/ddmut_ppi/api/list

POST - Job Submission

Arguments

- **pdb_accession** (optional) - 4 character PDB code
- **pdb_file** (optional) - file in [PDB format](#)
- **mutation_list** (required) - .txt or .csv file with mutation list. One mutation code per line (aaFrom + residueNumber + aaTo). Limited up to 500 mutations.
- **Reverse** (optional) - Whether to predict reverse mutations (True/False). Default is False

Return

- **job_id** - ID used for uniquely identify each job

Examples

- **curl**

```
$ curl https://biosig.lab.uq.edu.au/ddmut_ppi/api/list  
-X POST -i -F pdb_file=@/home/ubuntu/1cse.pdb -F  
mutations_list=@/home/ubuntu/mutations.txt  
HTTP/1.0 200 OK  
Content-Type: application/json  
Content-Length: 33  
Date: Tue, 21 Nov 2023 05:15:45 GMT  
{  
"job_id": "17005437412529573"  
}
```

GET - Retrieve Job Results

Arguments

- **job_id** - ID used for uniquely identify each job. Generated upon submission

Return

For jobs still being processed or waiting on queue, the message below will be returned from querying this endpoint:

- **message** - RUNNING

For jobs successfully processed by DDMut-PPI, the following data will be returned:

- Array list of results (in [json](#) format) for each mutation identified with a sequential identifying number

Examples

- **curl**

```
$ curl https://biosig.lab.uq.edu.au/ddmut_ppi/api/list  
-X GET -F job_id=16649535589333909  
{  
  "job_id": "16649535589333909",  
  "status": "DONE",  
  "results_page":  
  "http://10.50.192.76:5000/results_prediction/16649535589333909",  
  "I_L45A": {  
    "chain": "I",  
    "wild-type": "L",  
    "mutant": "A",  
    "position": "45",  
    "prediction": -1.97,  
    "outcome": "Decreasing"  
  },  
  "I_L45K": {  
    "chain": "I",  
    "wild-type": "L",  
    "mutant": "K",  
    "position": "45",  
    "prediction": -2.787,  
    "outcome": "Decreasing"  
  },  
  ...  
}
```

Alanine Scanning -

http://biosig.lab.uq.edu.au/ddmut_ppi/api/interface

POST - Job Submission

Arguments

- **pdb_accession** (optional) - 4 character PDB code
- **pdb_file** (optional) - file in [PDB format](#)

Return

- **job_id** - ID used for uniquely identify each job

Examples

- **curl**

```
$ curl https://biosig.lab.uq.edu.au/ddmut_ppi/api/interface -X POST -i -F pdb_file=@/home/ubuntu/1cse.pdb -F analysis_type='alascaan' HTTP/1.0 200 OK Content-Type: application/json Content-Length: 33 Date: Tue, 21 Nov 2023 05:10:59 GMT { "job_id": "16705716575422666" }
```

GET - Retrieve Job Results

Arguments

- **job_id** - ID used for uniquely identify each job. Generated upon submission

Return

For jobs still being processed or waiting on queue, the message below will be returned from querying this endpoint:

- **message** - RUNNING

For jobs successfully processed by DDMut-PPI, the following data will be returned:

- Array list of results (in [json](#) format) for each mutation identified with a

sequential identifying number

Examples

- **curl**

```
$ curl https://biosig.lab.uq.edu.au/ddmut_ppi/api/interface -X GET -F job_id=16705716575422666
{
  "job_id": "16705716575422666",
  "status": "DONE",
  "results_page":
    "http://10.50.192.76:5000/results_prediction/16705716575422666",
  "I_L45A": {
    "chain": "I",
    "wild-type": "L",
    "mutant": "A",
    "position": "45",
    "prediction": -1.97,
    "outcome": "Decreasing"
  },
  "I_L37A": {
    "chain": "I",
    "wild-type": "L",
    "mutant": "K",
    "position": "45",
    "prediction": -2.787,
    "outcome": "Decreasing"
  },
  ...
}
```

Saturation Mutagenesis -

http://biosig.lab.uq.edu.au/ddmut_ppi/api/interface

POST - Job Submission

Arguments

- **pdb_accession** (optional) - 4 character PDB code
- **pdb_file** (optional) - file in [PDB format](#)

Return

- **job_id** - ID used for uniquely identify each job

Examples

- **curl**

```
$ curl https://biosig.lab.uq.edu.au/ddmut_ppi/api/interface -X POST -i -F pdb_file=@/home/ubuntu/1cse.pdb -F analysis_type='satmut'
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 33
Date: Tue, 21 Nov 2023 05:10:59 GMT
{
  "job_id": "16705716575422666"
}
```

GET - Retrieve Job Results

Arguments

- **job_id** - ID used for uniquely identify each job. Generated upon submission

Return

For jobs still being processed or waiting on queue, the message below will be returned from querying this endpoint:

- **message** - RUNNING

For jobs successfully processed by DDMut-PPI, the following data will be returned:

- Array list of results (in **json** format) for each mutation identified with a sequential identifying number

Examples

- **curl**

```
$ curl https://biosig.lab.uq.edu.au/ddmut_ppi/api/interface -X GET -F job_id=16705716575422666
{
  "job_id": "16705716575422666",
  "status": "DONE",
}
```

```

"results_page":
"http://10.50.192.76:5000/results_prediction/16705716575422666",
"I_L45A": {
  "chain": "I",
  "wild-type": "L",
  "mutant": "A",
  "position": "45",
  "prediction": -1.97,
  "outcome": "Decreasing"
},
"I_L37A": {
  "chain": "I",
  "wild-type": "L",
  "mutant": "K",
  "position": "45",
  "prediction": -2.787,
  "outcome": "Decreasing"
},
...
}

```

Multiple Mutations: Manual Prediction -

https://biosig.lab.uq.edu.au/ddmut_ppi/api/manual

POST - Job Submission

Arguments

- **pdb_accession** (optional) - 4 character PDB code
- **pdb_file** (optional) - file in [PDB format](#)
- **mutations_list**(required) - Upload a plain text file with one multiple mutation per line. Limited up to 500 mutations. [Download sample](#)
- **email** (optional) - Email for contact when the job is finished

Return

- **job_id** - ID used for uniquely identify each job

Examples

- **curl**

```
$ curl https://biosig.lab.uq.edu.au/ddmut_ppi/api/manual -X
```



```
POST      -i      -F      pdb_accession="1cse"      -F
mutations_list=@/home/ubuntu/mutations.txt
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 33
Date: Wed, 22 Nov 2023 05:10:59 GMT
{
  "job_id": "160706007808"
}
```

GET - Retrieve Job Results

Arguments

- **job_id** - ID used for uniquely identify each job. Generated upon submission

Return

For jobs still being processed or waiting on queue, the message below will be returned from querying this endpoint:

- **message** - RUNNING

For jobs successfully processed by DDMut-PPI, the following data will be returned:

- **prediction** -

Examples

- **curl**
\$ curl https://biosig.lab.uq.edu.au/ddmut_ppi/api/manual -X GET -F job_id=160706007808
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 33
Date: Wed, 22 Nov 2023 05:13:29 GMT
{
 "E Q2M; I R48A; E T33A": {
 "average_distance": 22.14,
 "individual_predictions": {
 " E T33A": -0.71,
 " I R48A": -0.72,

```

"E Q2M": -0.163
},
"prediction": -1.27
},
"I D46A; I R48A": {
  "average_distance": 6.48,
  "individual_predictions": {
    " I R48A": -0.72,
    "I D46A": -2.287
  },
  "prediction": -2.29
},
"I D46A; I R48A; E T33A": {
  "average_distance": 11.29,
  "individual_predictions": {
    " E T33A": -0.71,
    " I R48A": -0.72,
    "I D46A": -2.287
  },
  "prediction": -2.89
},
"I D46A; I R48K": {
  "average_distance": 6.48,
  "individual_predictions": {
    " I R48K": -0.806,
    "I D46A": -2.287
  },
  "prediction": -2.27
}
}

```

Multiple Mutations: Systematic Evaluation -

https://biosig.lab.uq.edu.au/ddmut_ppi/api/systematic

POST - Job Submission

Arguments

- **pdb_accession** (optional) - 4 character PDB code
- **pdb_file** (optional) - file in [PDB format](#)
- **chain_id** (required) - Chain identifier for systematic evaluation of interface.

- **email** (optional) - Email for contact when the job is finished

Return

- **job_id** - ID used for uniquely identify each job

Examples

- **curl**

```
$ curl https://biosig.lab.uq.edu.au/ddmut_ppi/api/systematic
-X POST -i -F pdb_file=@/home/ubuntu/1cse.pdb -F chain="I"
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 33
Date: Wed, 22 Nov 2023 03:11:09 GMT
{
  "job_id": "160706129233"
}
```

GET - Retrieve Job Results

Arguments

- **job_id** - ID used for uniquely identify each job. Generated upon submission

Return

For jobs still being processed or waiting on queue, the message below will be returned from querying this endpoint:

- **message** - RUNNING

For jobs successfully processed by DDMut-PPI, the following data will be returned:

- Array list of results (in **json** format) for each mutation identified with a sequential identifying number

Examples

- **curl**

```
$ curl https://biosig.lab.uq.edu.au/ddmut_ppi/api/systematic
```

-X GET -F job_id=160706129233

HTTP/1.0 200 OK

Content-Type: application/json

Content-Length: 33

Date: Wed, 22 Nov 2023 08:01:12 GMT

{

"I T44P;I D46G;I L45G": {

"avg_distance": 4.39,

"individual_predictions": {

"D46G": -2.057,

"L45G": -2.019,

"T44P": -3.271

},

"prediction": -5.98

},

"I T44P;I D46G;I V43P": {

"avg_distance": 6.2,

"individual_predictions": {

"D46G": -2.057,

"T44P": -3.271,

"V43P": -1.899

},

"prediction": -6.01

},

"I T44P;I D46P;I L45D": {

"avg_distance": 4.39,

"individual_predictions": {

"D46P": -2.838,

"L45D": -2.257,

"T44P": -3.271

},

"prediction": -6.07

},

"I T44P;I D46P;I L45E": {

"avg_distance": 4.39,

"individual_predictions": {

"D46P": -2.838,

"L45E": -1.892,

"T44P": -3.271

},

"prediction": -5.94

```
},
"I T44P;I D46P;I L45G": {
  "avg_distance": 4.39,
  "individual_predictions": {
    "D46P": -2.838,
    "L45G": -2.019,
    "T44P": -3.271
  },
  "prediction": -5.93
},
"I T44P;I D46P;I V43P": {
  "avg_distance": 6.2,
  "individual_predictions": {
    "D46P": -2.838,
    "T44P": -3.271,
    "V43P": -1.899
  },
  "prediction": -6.09
},
"I T44P;I D46V;I L45E": {
  "avg_distance": 4.39,
  "individual_predictions": {
    "D46V": -1.992,
    "L45E": -1.892,
    "T44P": -3.271
  },
  "prediction": -5.94
},
"I T44P;I D46V;I V43P": {
  "avg_distance": 6.2,
  "individual_predictions": {
    "D46V": -1.992,
    "T44P": -3.271,
    "V43P": -1.899
  },
  "prediction": -6.06
},
"I T44P;I L45D;I D46G": {
  "avg_distance": 4.39,
  "individual_predictions": {
    "D46G": -2.057,
```

```
"L45D": -2.257,  
"T44P": -3.271  
,  
"prediction": -6.04  
,  
}
```

Biosig Lab [↗](#)

Our group is interested in developing and experimentally validating novel computational methods to exploit this data, enhancing the impact of genome sequencing, structural genomics, and functional genomics on biology and medicine.



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OPEN KNOWLEDGE

Best viewed using [Chrome](#) on 1280x960 resolution and above